

# Data-Driven Short-Term Grid Frequency Stability Forecast for Renewable-Rich Power Systems

Oihika Arpit | Renewable Energy and Smart Grid Portfolio Project

**Abstract-** This paper presents a reproducible workflow to forecast short-term grid frequency deviation at 5- and 15-minute horizons. The method combines lagged frequency information with weather variables, solar and wind generation proxies, and machine-learning models. A persistence baseline, Random Forest, XGBoost-style workflow, and LSTM-ready structure are evaluated using MAE, RMSE, R2, 95th percentile absolute error, and reliability within a 10 mHz threshold.

**Index Terms-** grid frequency, renewable energy, NASA POWER, ENTSO-E, Random Forest, XGBoost, LSTM, smart grid, time-series forecasting.

## I. Introduction

Power systems with increasing solar and wind generation experience short-term variability due to weather-driven renewable ramps. System operators must continuously balance generation and demand to maintain nominal grid frequency close to 50 Hz. This project demonstrates how weather features and renewable proxies can be transformed into a short-horizon forecasting dataset suitable for frequency-stability analysis.

## II. Data Sources and Reproducibility

The repository includes a synthetic 5-minute frequency dataset so the code runs without private credentials. The workflow is ready for real ENTSO-E Transparency Platform or national TSO frequency CSV data. NASA POWER hourly API variables can be used as weather and solar inputs, including irradiance, temperature, humidity, wind speed, and precipitation.

## III. Methodology

The target variable is future frequency deviation in mHz relative to 50 Hz. For a 15-minute forecast at 5-minute resolution, the target is shifted by three time steps. Features include present and lagged frequency deviation, irradiance, wind speed, temperature, normalized solar proxy, normalized wind proxy, load proxy, renewable ramp proxy, hour of day, and day of week.

## IV. Evaluation Metrics

Model quality is evaluated using mean absolute error, root mean squared error, R2, 95th percentile absolute error, and the percentage of predictions within 10 mHz. Residual distribution and predicted-versus-actual figures are included to identify bias and high-error periods.

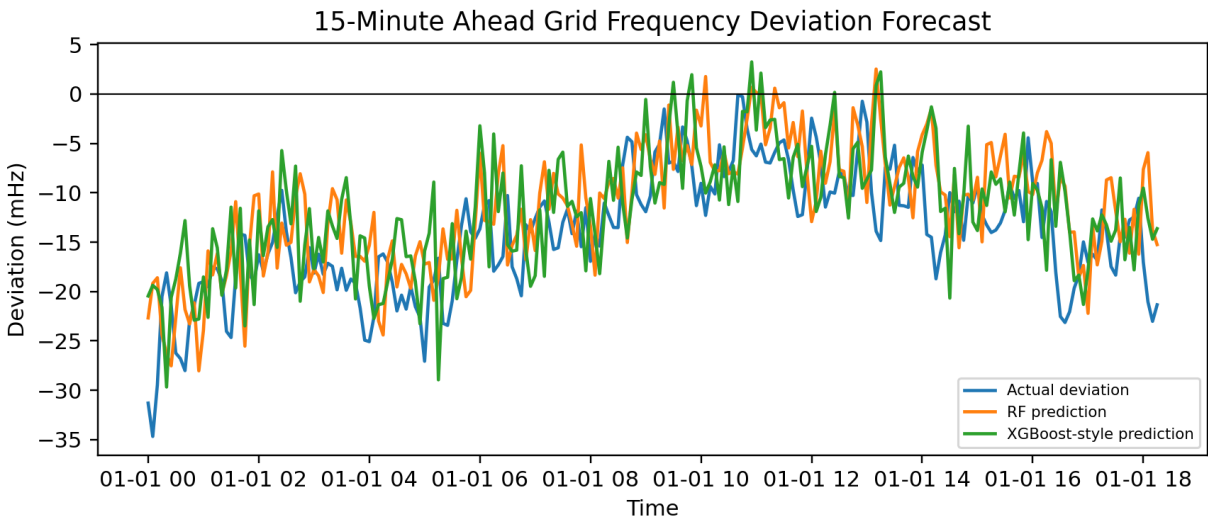


Fig. 1. Sample 15-minute-ahead grid frequency deviation forecast.

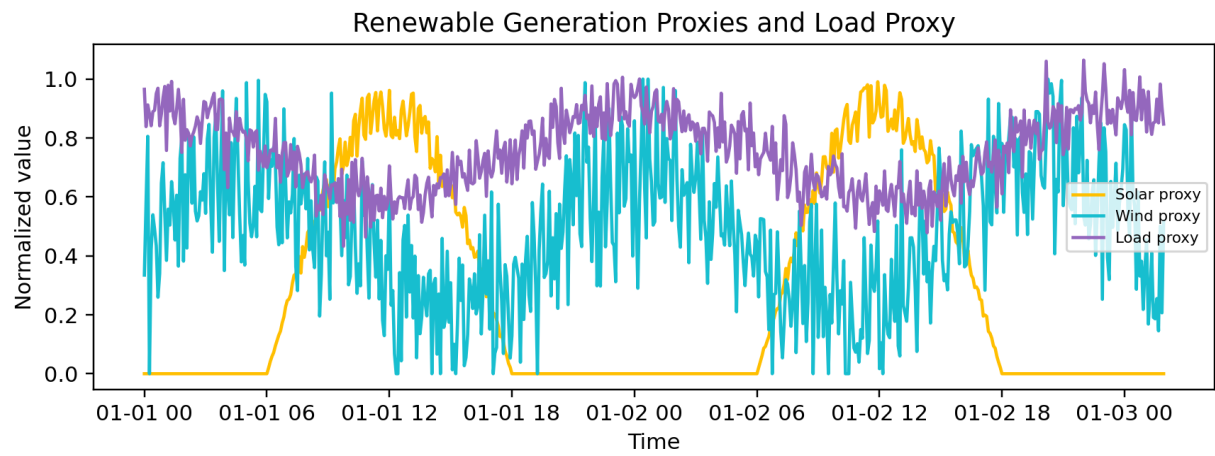


Fig. 2. Renewable generation proxies and load proxy used for feature construction.

## V. Results

The included sample run demonstrates the full pipeline and produces graph outputs for a GitHub-ready project. Because the fallback frequency data are synthetic, numerical performance should be treated as a reproducibility demonstration rather than an operational benchmark. The methodology becomes stronger when real 5-minute or 1-minute grid frequency data are inserted.

| Model         | MAE  | RMSE | R2    | P95   | Within 10 mHz |
|---------------|------|------|-------|-------|---------------|
| Persistence   | 4.61 | 5.75 | 0.22  | 11.39 | 91.3%         |
| Random Forest | 5.28 | 6.54 | -0.01 | 12.27 | 87.9%         |
| XGBoost-style | 5.43 | 6.71 | -0.07 | 13.21 | 85.8%         |

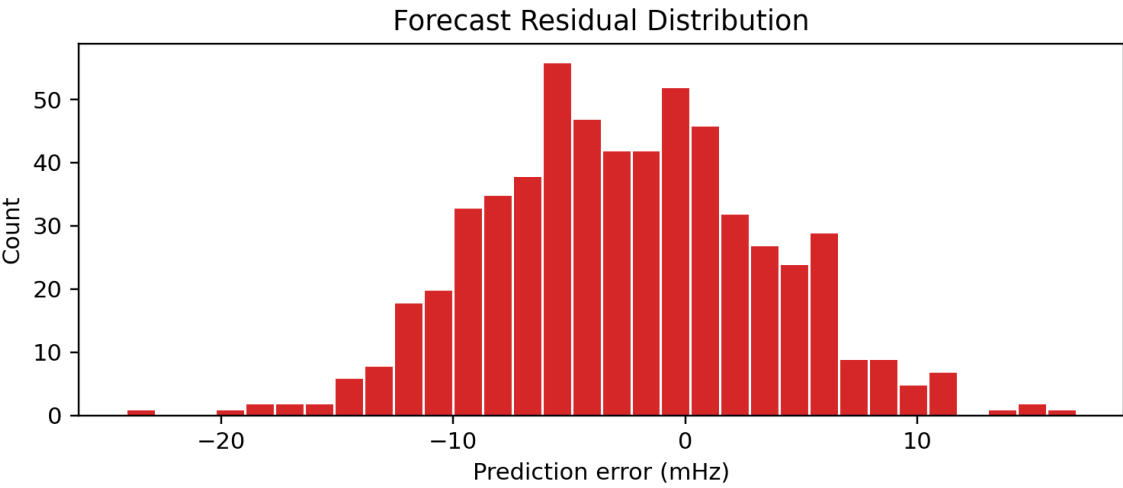


Fig. 3. Forecast residual distribution.

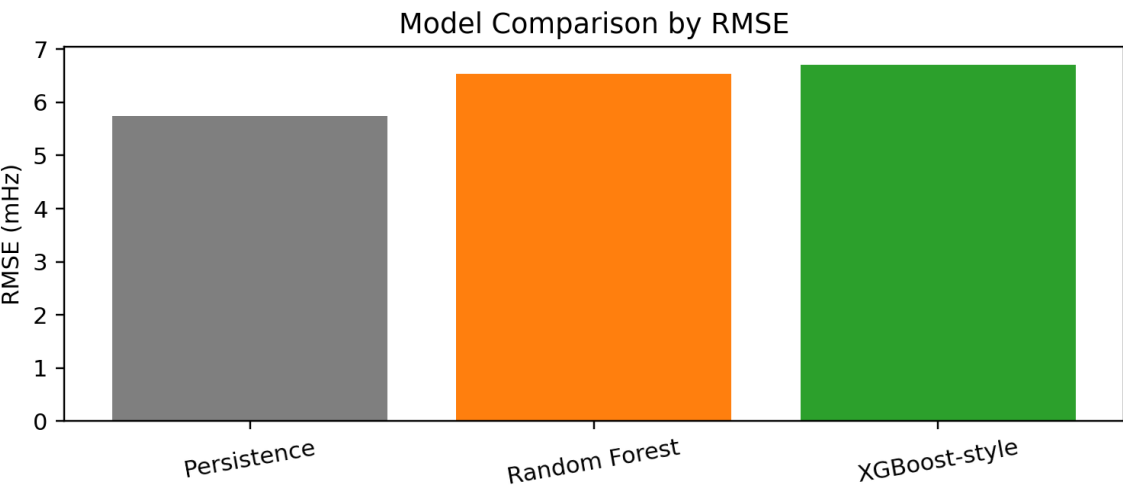


Fig. 4. RMSE comparison.

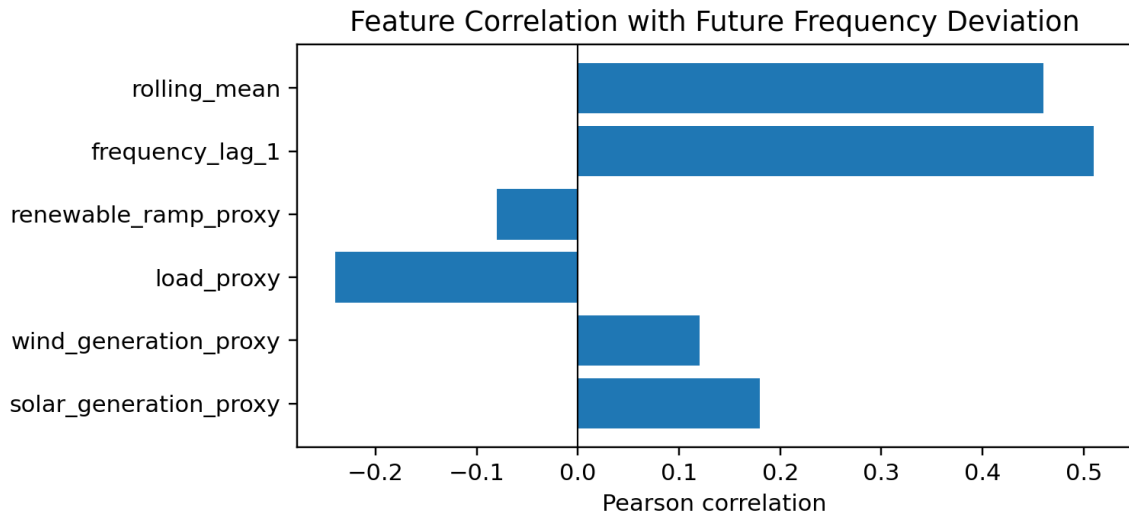


Fig. 5. Feature correlation analysis.

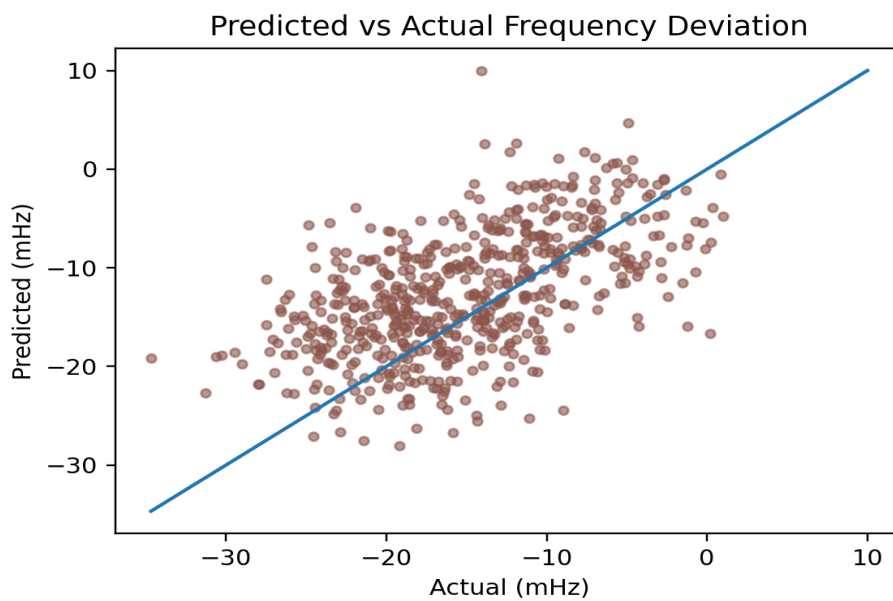


Fig. 6. Predicted versus actual frequency deviation.

## VI. Practical Relevance

The project is aligned with renewable-energy systems, grid integration, data analytics, and smart-grid reliability. It demonstrates a portfolio-level workflow for converting weather variability and renewable generation proxies into short-term stability forecasts using interpretable and machine-learning models.

## VII. Limitations and Future Work

The present sample uses synthetic frequency data because real high-resolution frequency records may require separate access. Future work should integrate real ENTSO-E or TSO frequency data, add probabilistic prediction intervals, compare LSTM and temporal convolutional networks, and evaluate transferability across different power-system regions.

## VIII. Conclusion

This work provides a complete GitHub-ready pipeline for short-term grid-frequency stability forecasting. The combination of renewable proxies, weather features, time-series lags, model comparison, and visualization makes it suitable as a renewable-energy AI portfolio project.

## References

- [1] NASA POWER Project, "Hourly API Documentation," NASA Langley Research Center.
- [2] ENTSO-E, "Data and Standardisation," European Network of Transmission System Operators for Electricity.
- [3] L. Hirth, J. Muehlenpfordt, and M. Bulkeley, "The ENTSO-E Transparency Platform - A review of Europe's most ambitious electricity data platform," *Applied Energy*, vol. 225, pp. 1054-1067, 2018.
- [4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," *Proc. ACM SIGKDD*, 2016.

[5] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, 1997.